Classless addressing

Given the network block 10.10.0.0/23, design an efficient IP addressing scheme using Variable Length Subnet Masking (VLSM).

The organization has the following subnet requirements:

LAN Requirements

Data Center: requires 200 hosts

Office LAN: requires 100 hosts

Data centre

$2^8 = 254 - 2$
$= 252$ ✓

mask: 32 − 8
= 24

host bits = 32−23
= 9

Max Range
10.10.0.0
|
10.10.00000000.0
10.10.00000001.255 − 10.10.1.255

10.10.0.00000000 − 10.10.0.0
10.10.0.00000001 − 10.10.0.1
|
10.10.0.11111110 − 10.10.0.254
10.10.0.11111111 − 10.10.0.255

Office LAN:

$2^7 = 128 - 2$
$= 126$ ✓

mask: 32 − 7
25

10.10.1.00000000 − 10.10.1.0
10.10.1.00000001 − 10.10.1.1
|
10.10.1.01111110 − 10.10.1.126
10.10.1.01111111 − 10.10.1.127

You are allocated the base network block **10.5.8.0/22**.

You must divide this block to accommodate the following four departments. The requirements are listed in no particular order:

- Department W: 120 hosts
- Department X: 500 hosts
- Department Y: 60 hosts
- Department Z: 200 hosts

Host bits = 32 − 22 = 10

Max Range: 10.5.00001000.00000000 − 10.5.8.0

10.5.00001011.11111111 − 10.5.11.255

Department X:

$2^9 - 2 = 510$ ✓

Mask: 32 − 9
= 23

10.5.00001000.00000000 — 10.5.8.0

10.5.00001000.00000001 — 10.5.8.1

10.5.00001001.11111110 — 10.5.9.254

10.5.00001001.11111111 — 10.5.9.255

10.5.8.0/23

Department Z:

$2^8 - 2 = 254$ ✓ 10.5.10.0/24

mask: 32 − 8
= 24

10.5.10.00000000 — 10.5.10.0
10.5.10.00000001 — 10.5.10.1

10.5.10.11111110 — 10.5.10.254
10.5.10.11111111 — 10.5.10.255

Department W

$2^7 - 2 = 126$ ✓

Mask: 32 − 7
= 25

10.5.11.0$\underbrace{0000000}$ — 10.5.11.0
10.5.11.00000001 — 10.5.11.1

10.5.11.01111110 — 10.5.11.126
10.5.11.01111111 — 10.5.11.127

10.5.11.0/25

Department Y

$2^6 - 2 = 62$ ✓

Mask: 32 − 6
26

10.5.11.10$\underbrace{000000}$ — 10.5.11.128

10.5.11.10000001 — 10.5.11.129

10.5.11.10111110 — 10.5.11.190
10.5.11.10111111 — 10.5.11.191

10.5.11.128/26

The Scenario:

You are given the Class C network address **192.168.100.0**.

Your organization requires you to divide this network to create at least **5 separate subnets** for different floors of a building.

$2^b \geq 5$
$2^3 \geq 5$

2. $b = 3$ 1. /24 4. $32 - 2 = 30$ 3. /27

Network part: $24 + 3$
27

#1

192.168.100.00000000 — 192.168.100.0
192.168.100.00000001 — 192.168.100.1

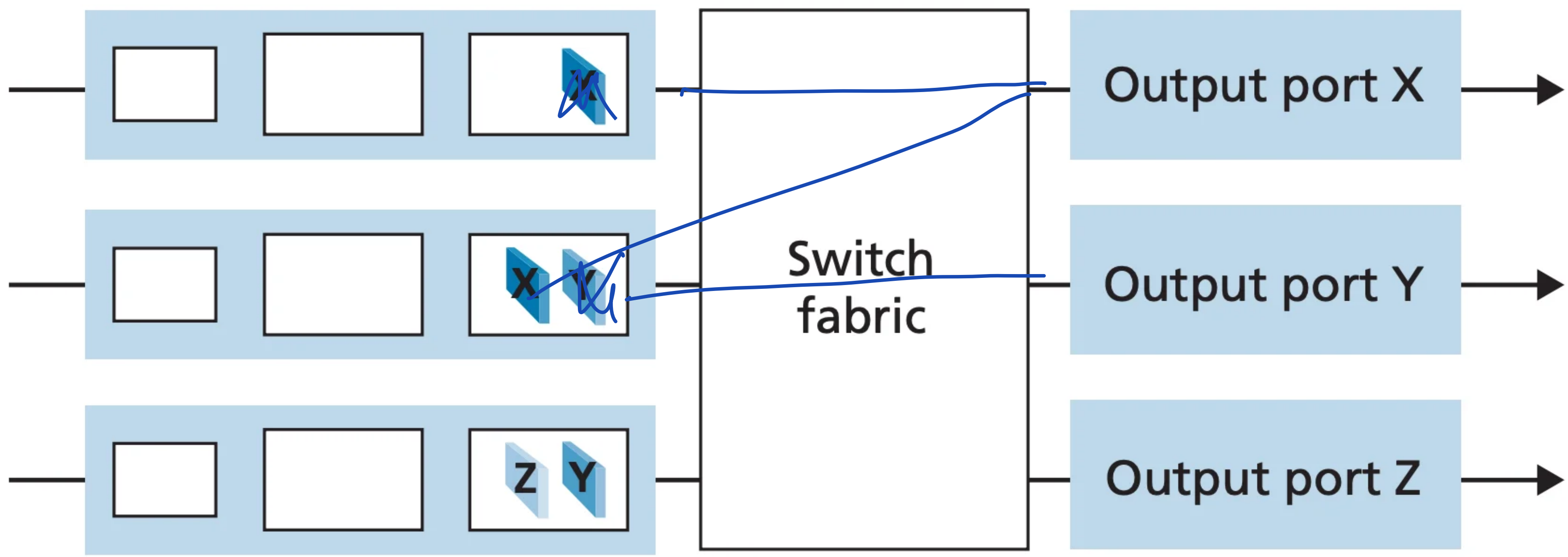
192.168.100.00011110 — 192.168.100.30
192.168.100.00011111 — 192.168.100.31

#2

192.168.100.32
192.168.100.33
|
192.168.100.62
192.168.100.63

⋮

P4:



X/Y X/Y Z

Sol

3

P5

$\frac{0.5}{0.25} : \frac{0.25}{0.25} : \frac{0.25}{0.25}$

2 : 1 : 1

(a) 11231123....

(b) 112112...

P8.

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1

11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

11100000 00	0
11100000 01000000	1
1110000	2
11100000 1	3
otherwise	3

Overflow:

Interface 2's prefix

11100000 00000000 00000000 00000000

interface 2's end:

11100001 01111111 11111111 11111111

+1

11100001 10000000 00000000 00000000

overflow start

11100001 11111111 11111111 11111111

overflow end

- **Interface 0 Range:**

  `11000000 00000000 00000000 00000000` through `11000000 00001111 11111111 11111111`

- **Interface 1 Range:**

  `11000000 00010000 00000000 00000000` through `11000000 00010000 11111111 11111111`

- **Interface 2 Range:**

  `11000000 00010001 00000000 00000000` through `11000000 00101111 11111111 11111111`

- **Otherwise:**

  Interface 3

11000000 0000	0
11000000 00010000	1
11000000 00	2
11000000 0011	3
otherwise	3

P14.

128.119.40.10000000 — 10.119.40.128

128.119.40.10111111 — 10.119.40.191

$2^b \geq 4$

$2^b \geq 2^2$

$b=2$

128.119.40.64/28 128.119.40.80/28

128.119.40.96/28 128.119.40.112/28

P15. Consider the topology shown in Figure 4.20. Denote the three subnets with hosts (starting clockwise at 12:00) as Networks A, B, and C. Denote the subnets without hosts as Networks D, E, and F.

a. Assign network addresses to each of these six subnets, with the following constraints: All addresses must be allocated from 214.97.254/23; Subnet A should have enough addresses to support 250 interfaces; Subnet B should have enough addresses to support 120 interfaces; and Subnet C should have enough addresses to support 120 interfaces. Of course, subnets D, E and F should each be able to support two interfaces. For each subnet, the assignment should take the form a.b.c.d/x or a.b.c.d/x – e.f.g.h/y.

b. Using your answer to part (a), provide the forwarding tables (using longest prefix matching) for each of the three routers.

subnet A:

$2^8 - 2 = 254$

Mask: 32 − 8
= 24

214.97.1111111 0. ...

214.97.255.255

$2^2 - 2 = 2$

214.97.254.0000 0000 – 24.97.254.0
214.97.254.00000001 – 24.97.254.1

|

214.97.254.11111110 – 24.97.254.254
214.97.254.11111111 – 24.97.254.255

subnet B:

$2^7 - 2 = 126$

Mask: 32 − 7
= 25

214.97.255.0000 0000 – 24.97.255.0
214.97.255.00000001 – 24.97.255.1

|

|

214.97.255.01111110 – 24.97.255.126
214.97.255.01111111 – 24.97.255.127

subnet C:

$2^7 - 2 = 126$

Mask: 32–7
: 25

214.97.255.10000000 – 24.97.255.128
214.97.255.10000001 – 24.97.255.128

|

214.97.255.11111110 – 24.97.255.254
214.97.255.11111111 – 24.97.255.255

subnet D:

$2^2 - 2 = 2$
curve up subnet A

mask: 32–2
= 30

Since A needs only 252 addresses, remaining 4 are free. so, we use them. Thus, A's end is now

24.97.254.251

∴ D starts:

24.97.254.252

|

24.97.254.255

subnet E:

$2^2 - 2 = 4$ curve up subnet B

mask: $32 - 2$
: 30

subnet B's end is:

24.97.255.123 → can't be 24.97.255.121

& E starts

24.97.255.124 → can't be 24.97.255.122
|
24.97.255.127

24.97.255.127

↳ because allocation must be in 2's power. It can't be 6. It needs to be 4 or 8, 4 is what we need.

subnet F

$2^2 - 2 = 4$

curve up subnet C

mask: $32 - 2$
: 30

subnet C's end is:

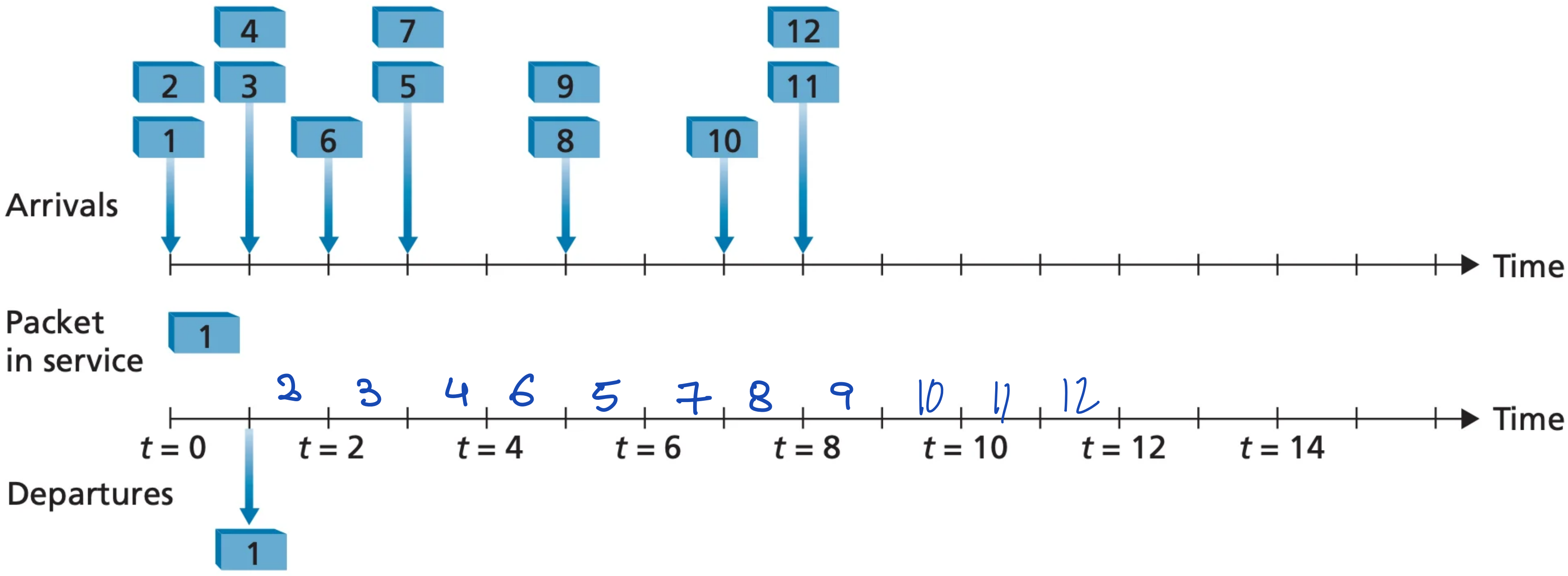
24.97.255.251

& E starts

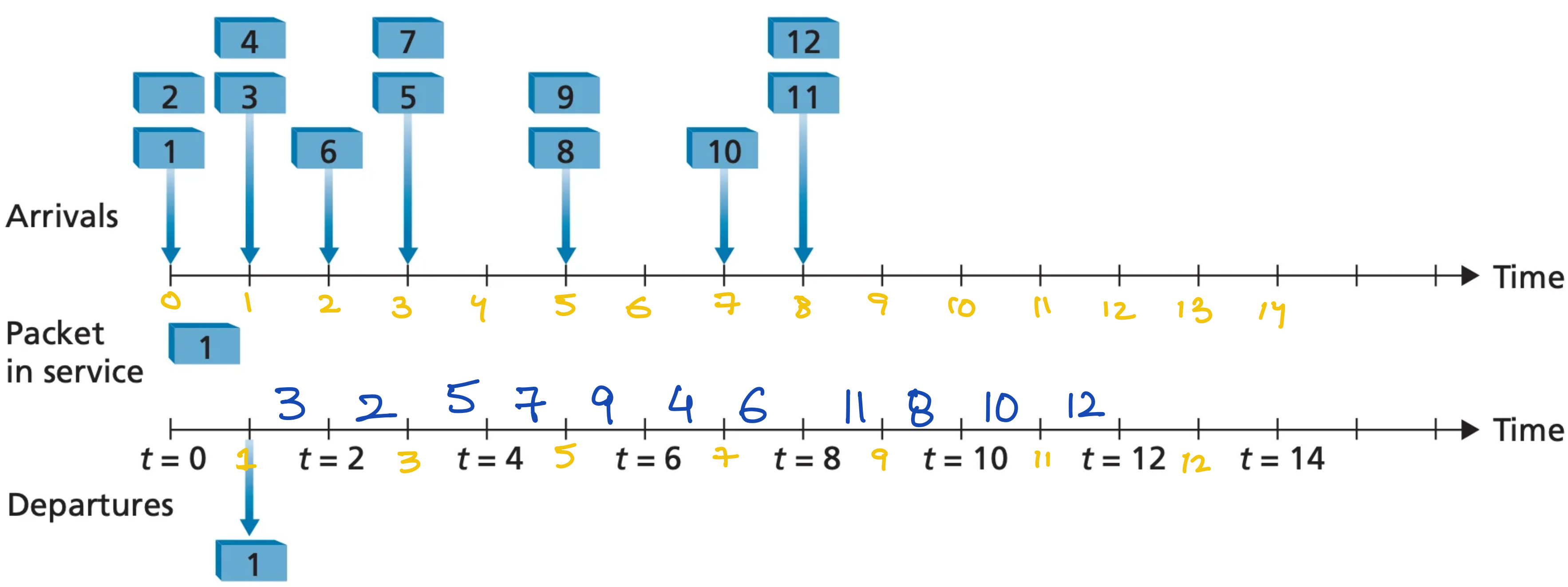
24.97.255.252
|
24.97.255.255

P6

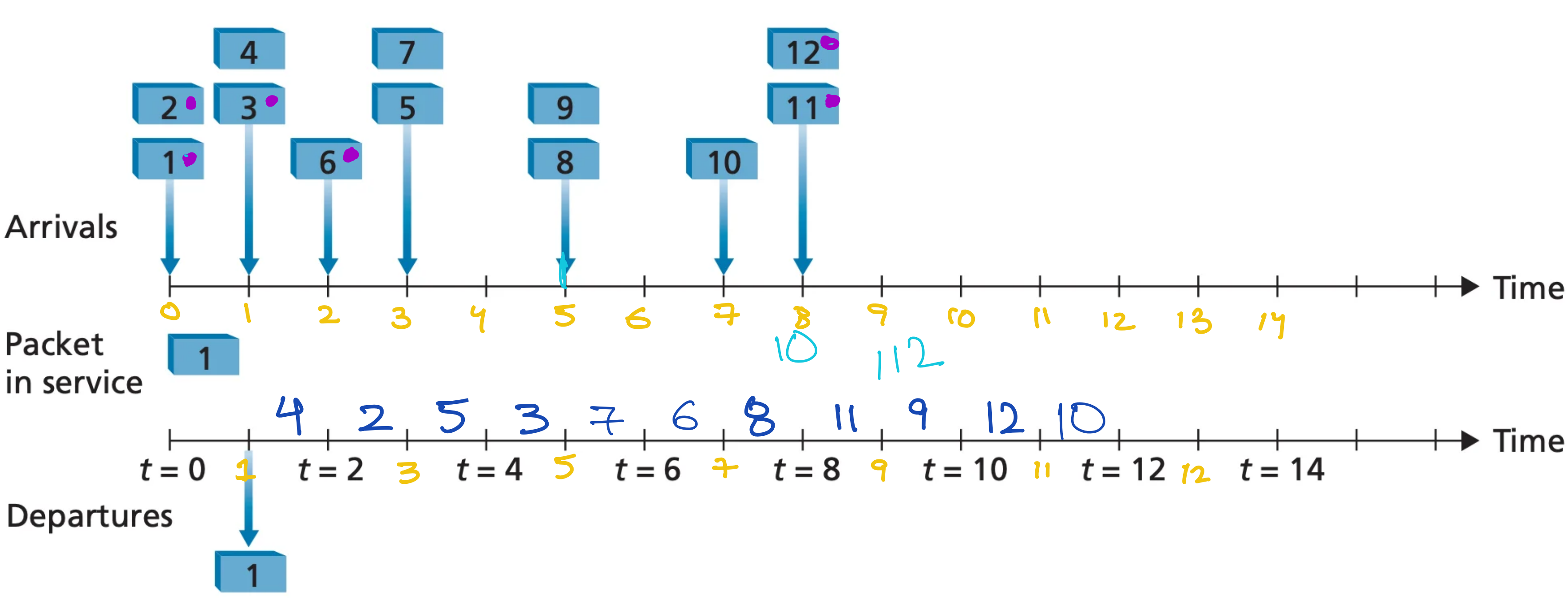
(a)



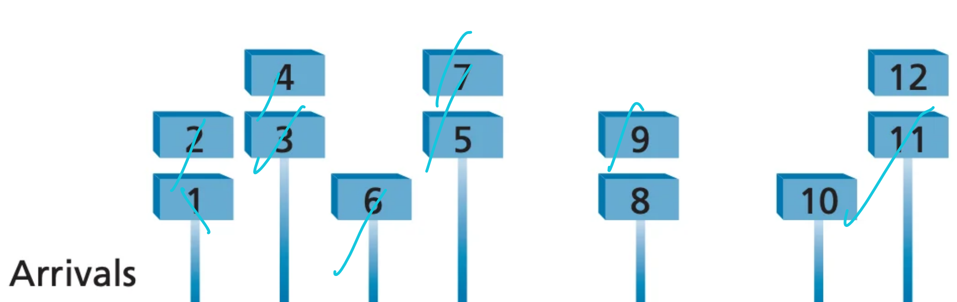
(b)



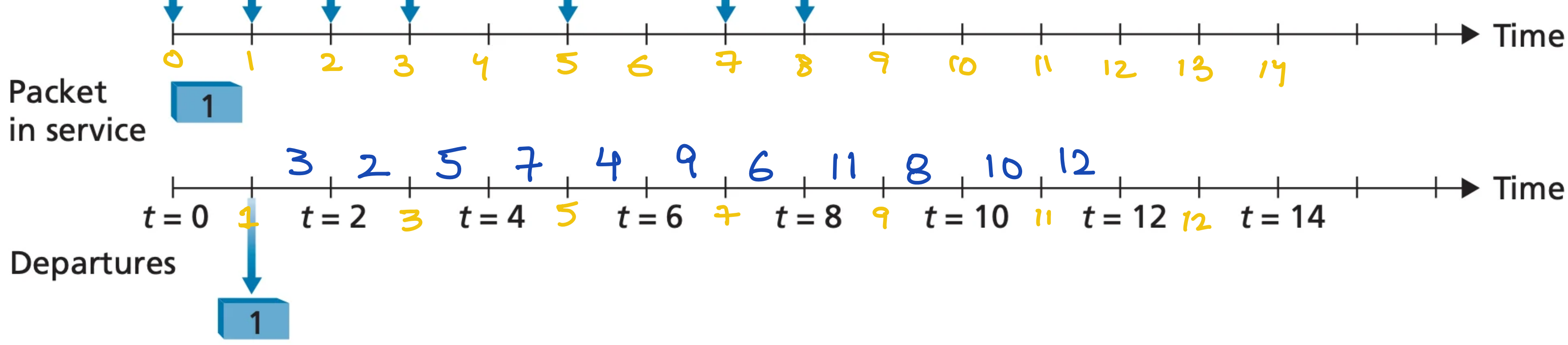
(c)



d)



ODD: 2
EVEN: 1



P8. Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as follows:

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

11100000	00	0
11100000	01000000	1
1110000		2
11100001	1	3
otherwise		3

The Scenario

You are the network architect for a rapidly expanding enterprise. Your company has been allocated a single Class B network block: `172.24.0.0/16`.

You must divide this space to accommodate the requirements of two different regional offices, ensuring optimal address utilization with zero overlapping blocks.

Region A (CIDR / VLSM Design)

Region A requires highly optimized allocations based on the exact host counts for four distinct departments. You must allocate these subnets sequentially, starting from the very beginning of the allocated block, assigning the largest subnets first to maintain binary boundary alignment.

- **Department 1:** 4,000 hosts
- **Department 2:** 2,000 hosts
- **Department 3:** 500 hosts
- **Department 4:** 60 hosts
- **WAN Links:** Two point-to-point router links (2 hosts required per link)

Department 01:

$$2^n - 2 = 2^{12} - 2$$
$$= 4094 \le 4000$$

172.24.00000000.00000000 — 172.24.0.0
172.24.00000000.00000001 — 172.24.0.1

|

172.24.00001111.11111110 — 172.24.15.254
172.24.00001111.11111111 — 172.24.15.255

172.24.0.0/20

Department 02:

$$2^n - 2 = 2^{11} - 2$$
$$= 2046 \le 2000$$

172.24.00010000.00000000 — 172.24.16.0
172.24.00010000.00000001 — 172.24.16.1

|

172.24.00010111.11111110 — 172.24.23.254

172.24.00010111.11111111 – 172.24.23.255

172.24.16.0/21

Department 03

$2^n - 2 = 2^9 - 2$
$= 510 \leq 500$

172.24.00011000.00000000 – 172.24.24.0
172.24.00011000.00000001 – 172.24.24.1

172.24.00011001.11111110 – 172.24.25.254
172.24.00011001.11111111 – 172.24.25.255

172.24.24.0/23

Department 04:

$2^n - 2 = 2^6 - 2$
$= 62$

172.24.00011010.00000000 – 172.24.26.0
172.24.00011010.00000001 – 172.24.26.1

172.24.00011010.00111110 – 172.24.26.62
172.24.00011010.00111111 – 172.24.26.63

172.24.26.0/26

WAN Link 1:

$2^2 - 2 = 2 \leq 2$

172.24.00011010.01000000 – 172.24.26.64
172.24.00011010.01000001 – 172.24.26.65

172.24.00011010.01000010 — 172.24.26.66
172.24.00011010.01000011 — 172.24.26.67

172.24.26.64/30

WAN Link 2:

$2^2 - 2 = 2 \leq 2$

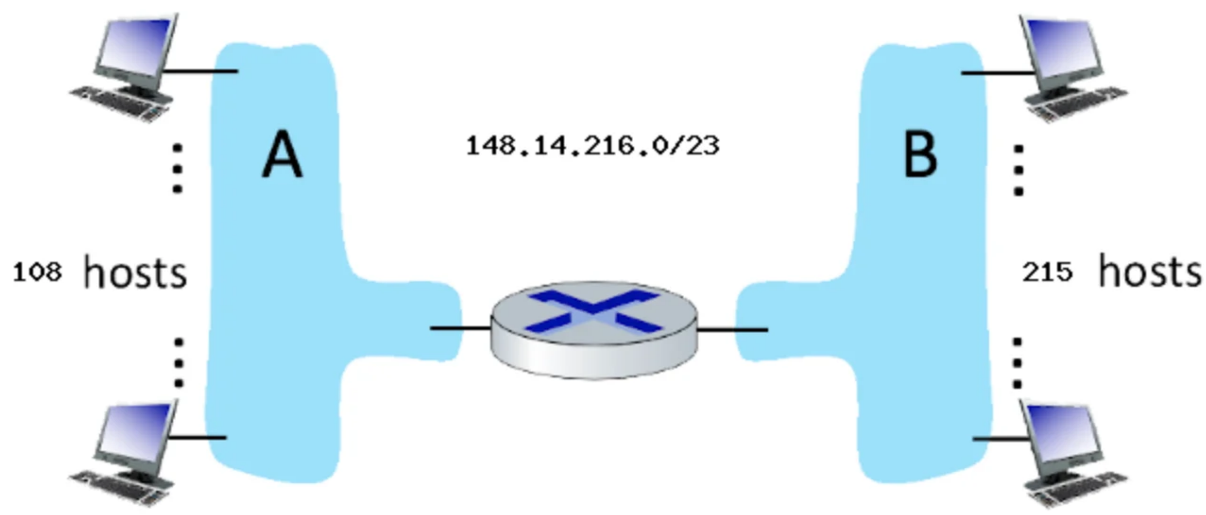
172.24.00011010.01000100 — 172.24.26.68
172.24.00011010.01000101 — 172.24.26.69

172.24.00011010.01000110 — 172.24.26.70
172.24.00011010.01000111 — 172.24.26.71

172.24.26.68/30

SUBNET ADDRESSING

Consider the router and the two attached subnets below (A and B). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 148.14.216.0/23



B:

$2^n - 2 = 2^8 - 2$

$= 254 \le 215$

148.14.216.00000000

|

148.14.216.11111111

A:

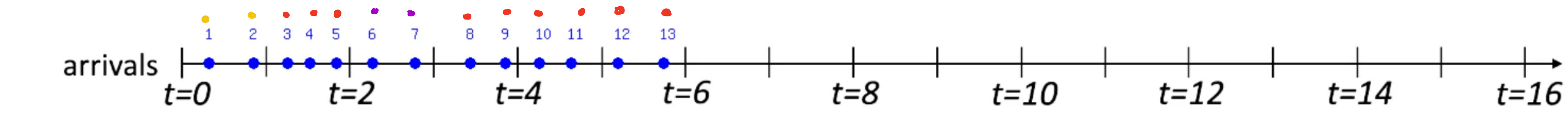
$2^n - 2 = 2^7 - 2$

$= 126 \le 108$

148.14.217.00000000

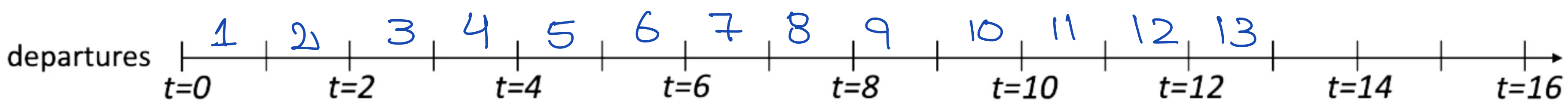
|

148.14.217.01111111

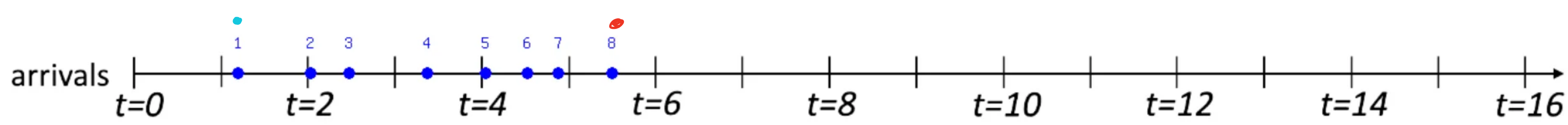


Packets (#: time): 1: 0.33, 2: 0.86, 3: 1.26, 4: 1.52, 5: 1.83, 6: 2.26, 7: 2.75, 8: 3.41, 9: 3.82, 10: 4.22, 11: 4.59, 12: 5.14, 13: 5.68

Packets (#: class): 1: 1 , 2: 1 , 3: 2 , 4: 2 , 5: 2 , 6: 3 , 7: 3 , 8: 2 , 9: 2 , 10: 2 , 11: 2 , 12: 2 , 13: 2

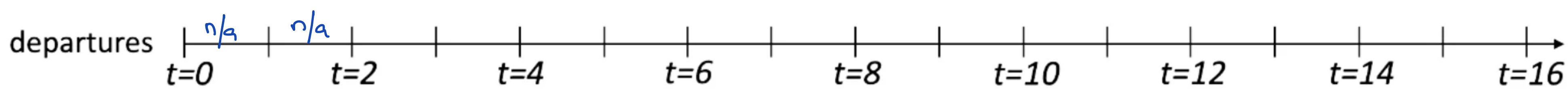


1: 5
2: 3
3: 2



Packets (#: time): 1: 1.19, 2: 2.02, 3: 2.45, 4: 3.34, 5: 4.01, 6: 4.48, 7: 4.83, 8: 5.44

Packets (#: class): 1: 2 , 2: 3 , 3: 3 , 4: 3 , 5: 3 , 6: 3 , 7: 3 , 8: 1



C_1 C_2 C_3

1: ~~8~~

2: ~~1~~

3: ~~2~~, ~~3~~, ~~4~~, 5, 6, 7

$t=0$: n/a

$t=1$: n/a

$t=2$: C_1 C_2: 1

$t=3$: C_2, C_3 2

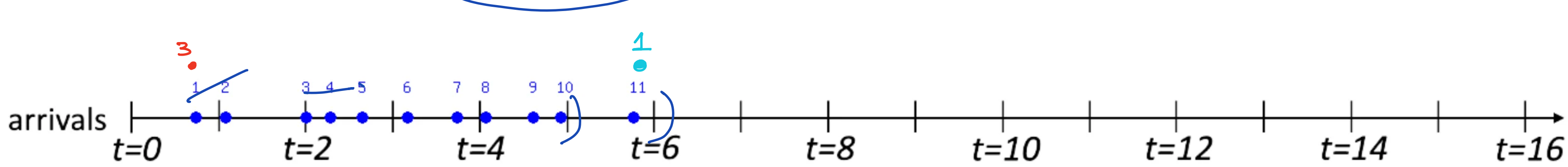
$t=4$: C_3, 3

$t=5$: C_1, 4

$t=6$: C_2: 5

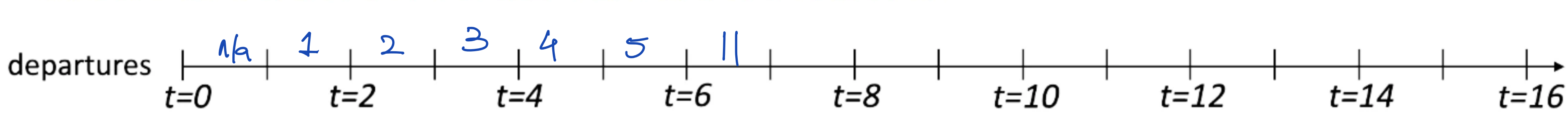
$t=7$: 6

$C_1 \to C_2 \to C_3$

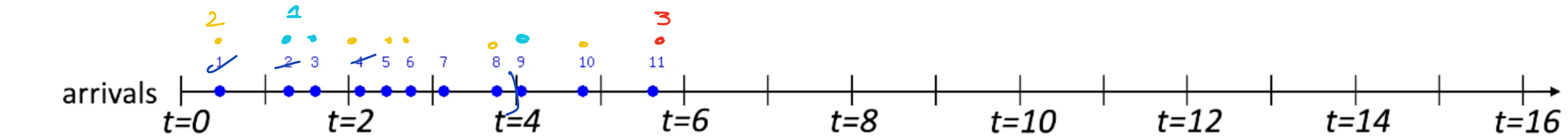


Packets (#: time): 1: 0.74, 2: 1.08, 3: 1.99, 4: 2.27, 5: 2.63, 6: 3.14, 7: 3.71, 8: 4.03, 9: 4.57, 10: 4.88, 11: 5.71

Packets (#: class): 1: 3 , 2: 2 , 3: 2 , 4: 2 , 5: 2 , 6: 2 , 7: 2 , 8: 2 , 9: 2 , 10: 2 , 11: 1

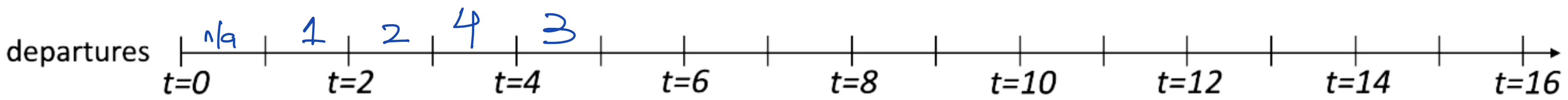


$C_1 \to C_2 \to C_3$



Packets (#: time): 1: 0.47, 2: 1.28, 3: 1.59, 4: 2.12, 5: 2.43, 6: 2.72, 7: 3.1, 8: 3.73, 9: 4.02, 10: 4.74, 11: 5.57

Packets (#: class): 1: 2 , 2: 1 , 3: 1 , 4: 2 , 5: 2 , 6: 2 , 7: 2 , 8: 2 , 9: 1 , 10: 1 , 11: 3



1: ~~2~~, 3 9 10

2: ~~1~~ 4 5 6 7 8

3: ~~11~~

0.55 : 0.45

1.22 : 2 1

11 : 9

t=0 na

t=1 c1 c2 1

t=2 c2 c1 ~~2~~

t=3 c3 c1 3

t=4 c1 c2 ~~4~~

t=5 c2 5

t=6 c3 11

c1 → c2